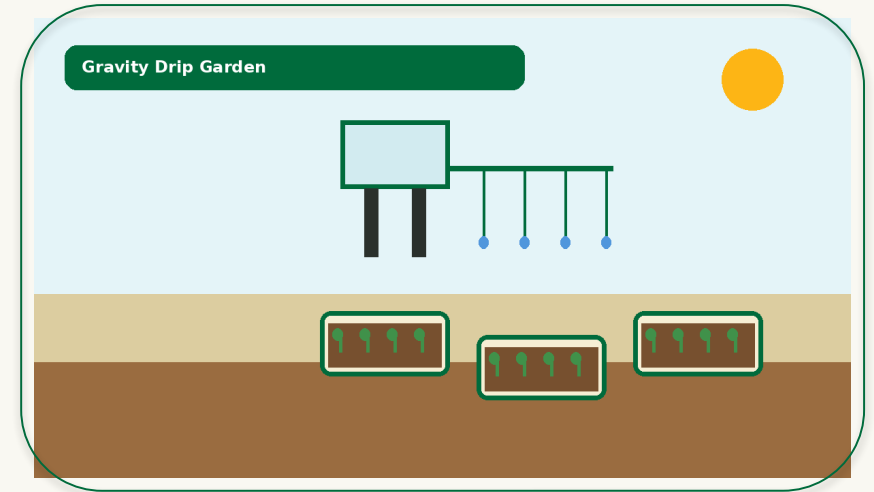


Gravity Drip Garden

Introduction to Irrigation Science

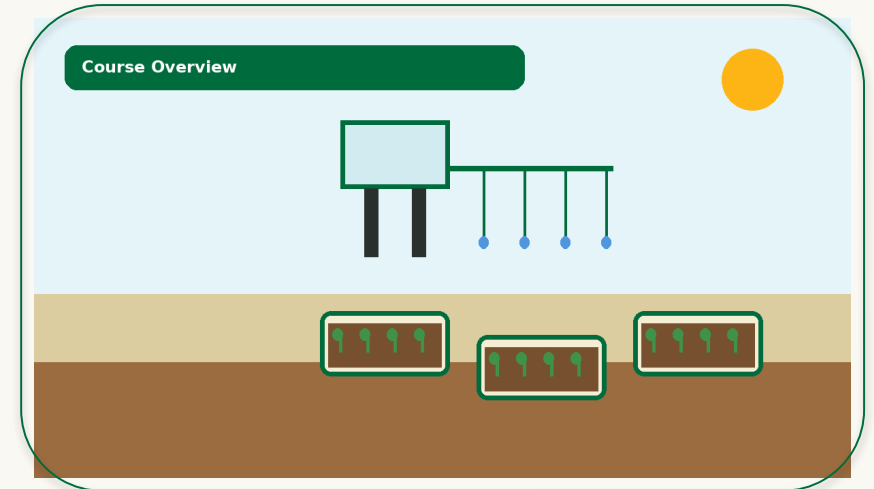
- LIFEWS AgriShine™ GrowMeal™ Primary 4-6 Teacher Training
- Audience: Primary 4-6 teachers, teaching assistants and garden facilitators
- Format: Self-paced online course + supervised garden practical
- Recommended duration: 3 hours online + 90 minutes practical
- Learning progression: measure -> experiment -> record -> analyze -> improve



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the system and its educational purpose; teach all ten technical topics at Primary 4-6 level; guide safe practical work and measurement; connect science, mathematics and agriculture; maintain useful records; help pupils interpret observations without overclaiming.
- Core materials: teacher guide, system diagram, child-safe tools, labels, clipboards, rulers/tapes, graduated containers or measuring devices where relevant, observation/data sheets, clean harvest containers, handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical lesson, and submit a short lesson, measurement or investigation plan.



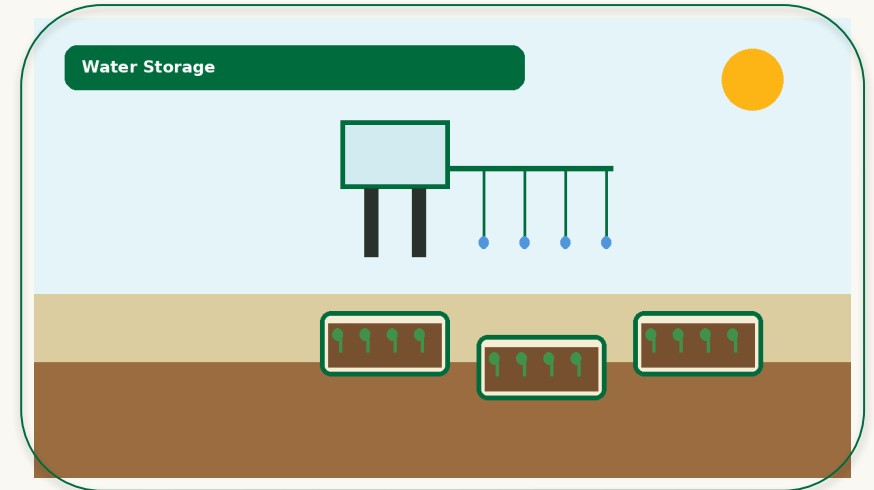
TEACHER PRACTICE

- Training sequence: EXPLAIN -> DEMONSTRATE -> MEASURE -> RECORD -> INTERPRET.

Water Storage

Teacher knowledge and practical application

- Identify the storage tank/reservoir as the starting point of the gravity-drip system.
- Explain why stored water should be protected from contamination and uncontrolled access.
- Relate tank volume to the amount of water available for irrigation.
- Use visible level indicators where installed to support measurement activities.
- Keep pupils away from climbing, open tanks and structural supports.



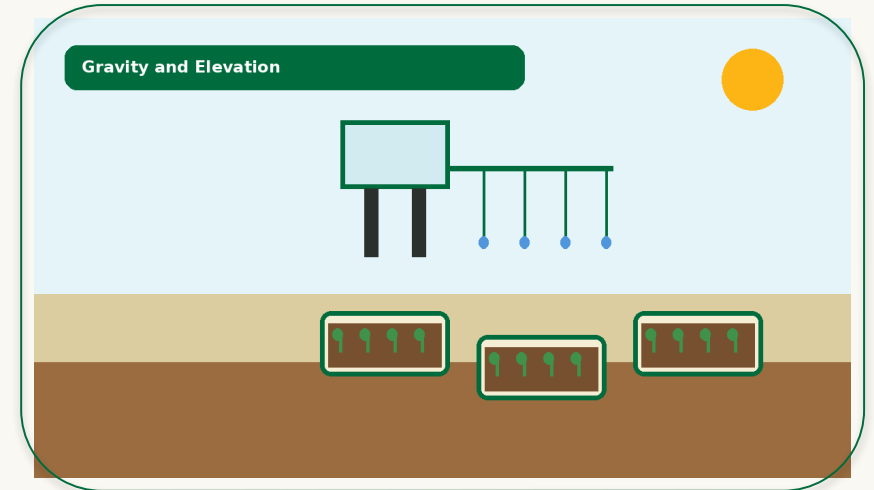
TEACHER PRACTICE

- Teacher task: trace the system from storage tank to one crop bed.

Gravity and Elevation

Teacher knowledge and practical application

- Explain that water at a higher elevation can move downward through pipes because of gravity.
- Compare a raised tank with a container at ground level using a safe demonstration.
- Introduce elevation as vertical height relative to the outlet or crop area.
- Avoid oversimplifying: pipe losses and system design also affect delivery.
- Use the installed system rather than asking pupils to lift heavy water containers.



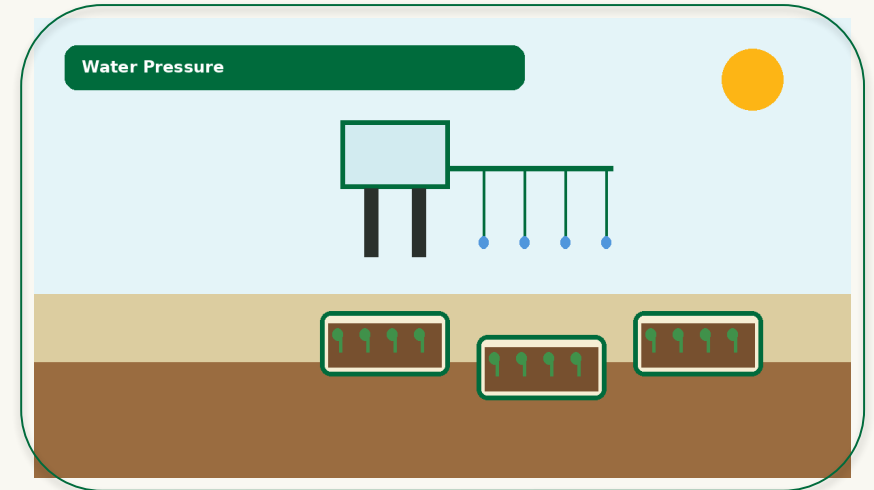
TEACHER PRACTICE

- Prepare a safe high-versus-low water-flow demonstration.

Water Pressure

Teacher knowledge and practical application

- Introduce pressure as the force that helps move water through the irrigation network.
- Explain that greater water height can increase pressure in a gravity-fed system.
- Use simple pressure observations rather than advanced equations at this level.
- Compare emitter behavior at different points without adjusting unsafe fittings.
- Discuss why excessive or insufficient pressure can affect irrigation uniformity.



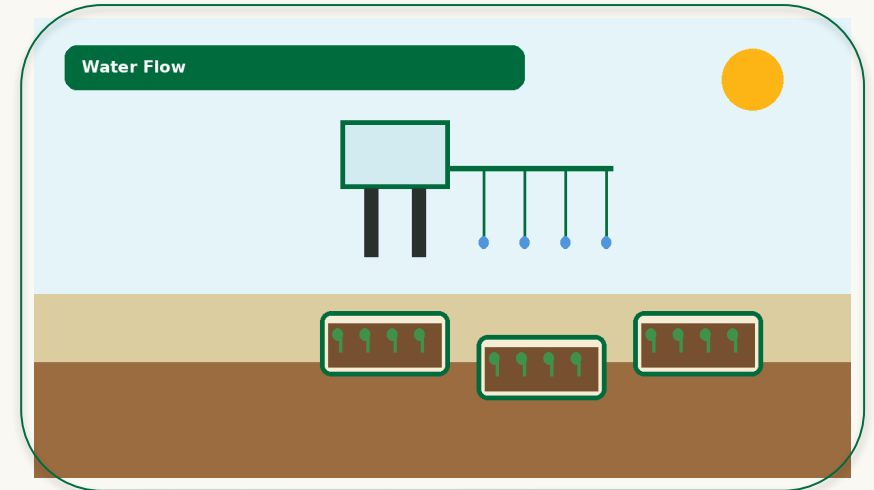
TEACHER PRACTICE

- Identify observable signs of low or uneven pressure.

Water Flow

Teacher knowledge and practical application

- Define flow as the amount of water moving through the system over time.
- Observe water moving through a transparent section, outlet or measured container where safe.
- Measure simple flow by collecting water for a fixed time.
- Record volume and time together.
- Explain that blocked pipes, valves and elevation can change flow.



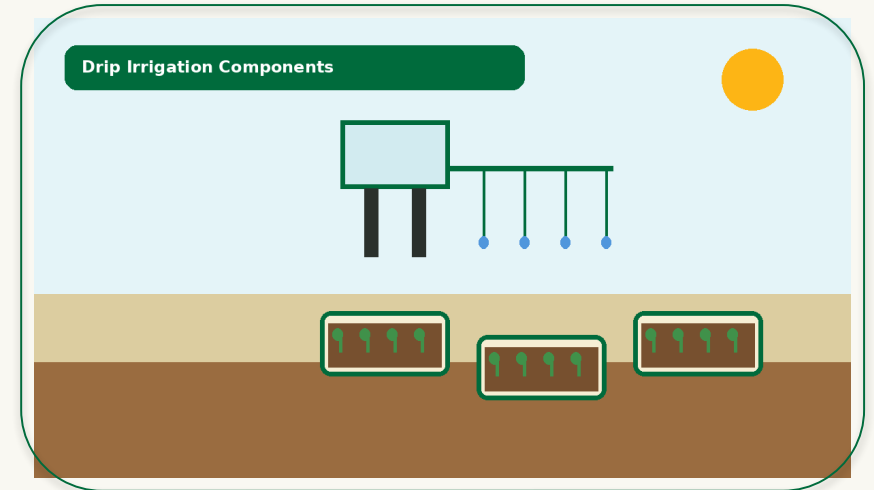
TEACHER PRACTICE

- Prepare a one-minute flow measurement activity.

Drip Irrigation Components

Teacher knowledge and practical application

- Identify tank, outlet, filter, mainline, laterals, valves, connectors and emitters.
- Explain the function of each component in the water path.
- Use a physical component board or installed system for identification.
- Keep pupils from dismantling pressurized or fixed components.
- Teach correct names so maintenance reports are clear.



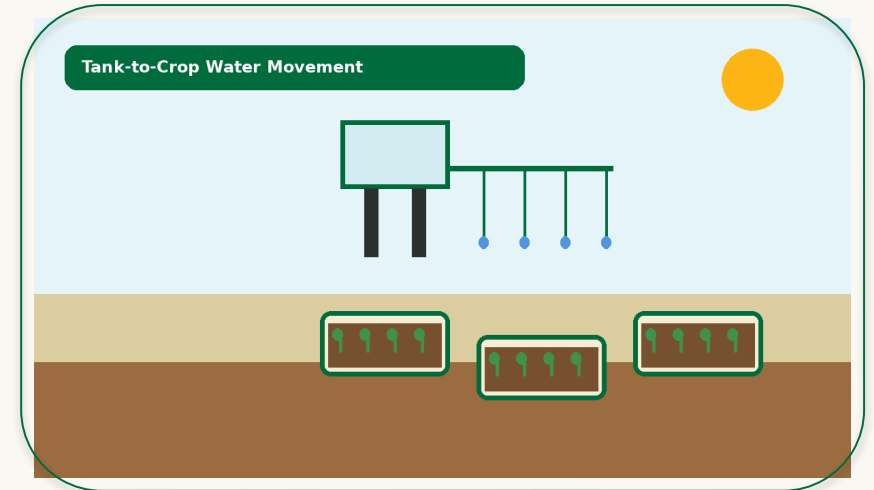
TEACHER PRACTICE

- Create a component-identification station.

Tank-to-Crop Water Movement

Teacher knowledge and practical application

- Trace water from storage through the mainline and laterals to emitters and root zones.
- Use arrows on a simple system diagram.
- Observe where water enters the soil and how the wet area develops.
- Discuss why irrigation targets the root zone.
- Identify points where water can be lost through leaks or overflow.



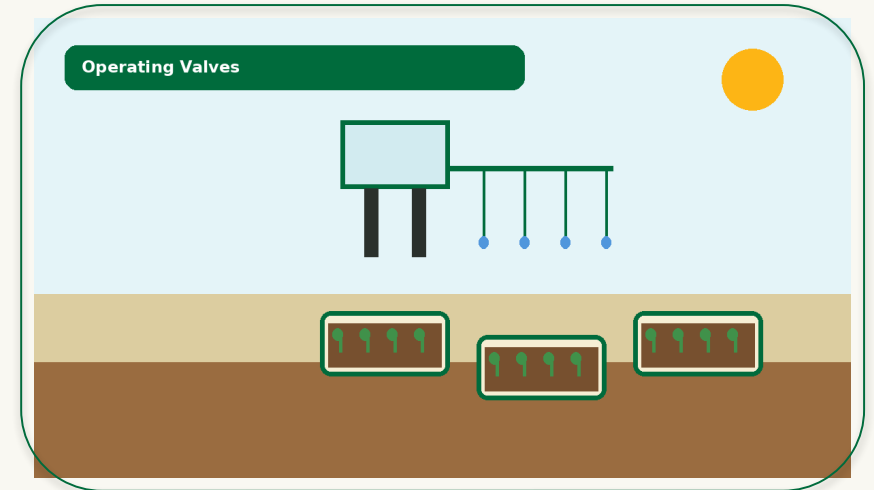
TEACHER PRACTICE

- Prepare a tank-to-root-zone mapping activity.

Operating Valves

Teacher knowledge and practical application

- Explain valves as devices used to start, stop or direct water flow.
- Teachers demonstrate the correct open/closed positions on approved low-risk valves.
- Operate valves slowly to avoid sudden changes in the system.
- Label irrigation zones where possible.
- Pupils should not operate pump controls, electrical equipment or restricted valves.



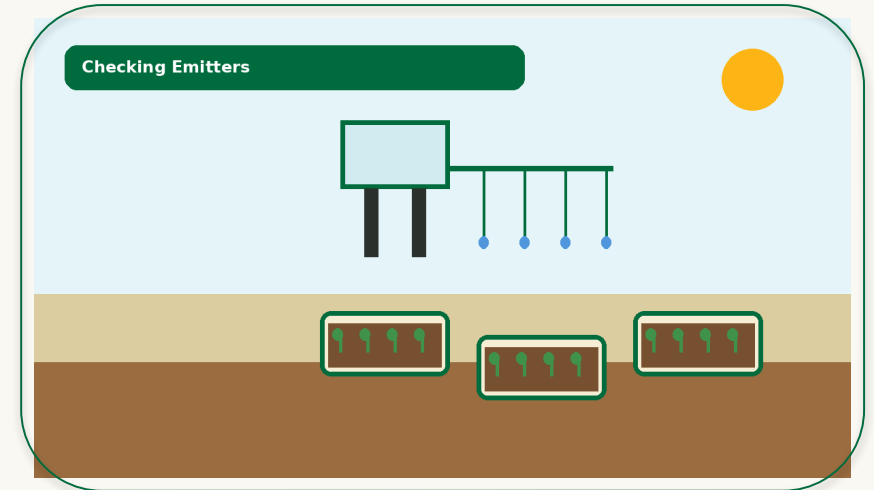
TEACHER PRACTICE

- Demonstrate a safe start-up and shutdown sequence.

Checking Emitters

Teacher knowledge and practical application

- Inspect emitters for visible flow, blockage, damage or disconnection.
- Compare several emitters in the same zone.
- Clean or replace blocked emitters only according to the maintenance procedure.
- Avoid inserting sharp objects into emitters during pupil activities.
- Record emitter problems by bed or line.



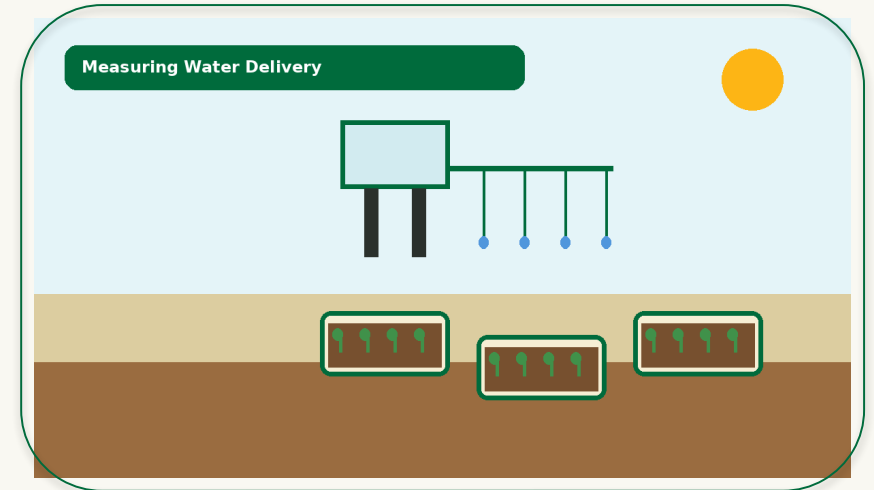
TEACHER PRACTICE

- Prepare an emitter inspection checklist.

Measuring Water Delivery

Teacher knowledge and practical application

- Collect emitter discharge in equal containers for the same amount of time.
- Measure delivered volume using graduated containers.
- Compare values to identify large differences.
- Repeat questionable measurements before concluding that an emitter is faulty.
- Use results to discuss irrigation uniformity in simple terms.



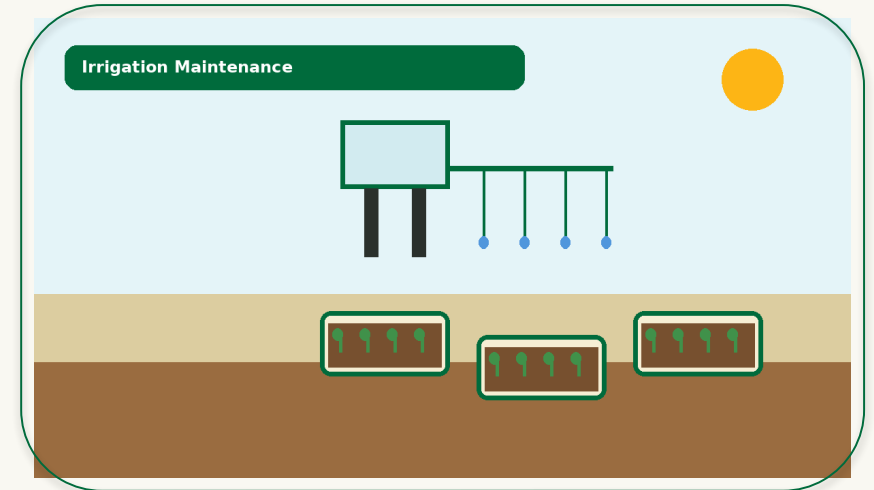
TEACHER PRACTICE

- Measure and compare at least four emitters.

Irrigation Maintenance

Teacher knowledge and practical application

- Inspect tank level, filter, pipes, connectors, valves and emitters routinely.
- Repair leaks promptly and keep lines protected from trampling.
- Flush or clean components according to system guidance.
- Maintain dry, safe pathways and secure the tank structure.
- Keep a maintenance record with date, problem and action taken.



TEACHER PRACTICE

- Completion task: create a gravity-drip inspection and maintenance routine.